

ESS Investigation

Weather and Climate

The Sky is the Limit — Decoding Weather Conditions

Teacher Materials

Investigation Overview

In this investigation, students will evaluate and communicate meteorological information to explain how the movement and the interactions of air masses result in changes in weather conditions. Students will develop and use models, weather maps, data, and satellite images to construct explanations, analyze and interpret data, and use math and computational thinking to describe atmospheric conditions and the resulting weather phenomena.

In **Session 1: Collaborative Student Work**, students will collaborate in pairs to individually construct and use a model of atmospheric pressure in columns of warm and cold air, plot the path of a hurricane, and use a sequence of weather maps to describe weather conditions.

In **Part 1**, students will:

- construct a model of a column of warm air and a column of cold air by using “pressure blocks” in order to explain how variations in air temperature cause differences in density and air pressure
- explain how the differences between warm and cold air parcels produce horizontal variations in air pressure at different altitudes in the atmosphere
- plot the path of either a northern or southern hemisphere hurricane in order to evaluate and communicate how global prevailing wind patterns affect weather phenomena in different regions of the world

In **Part 2**, students will continue to collaborate in pairs using a sequence of weather maps to:

- identify and draw the location of both a warm front and a cold front on a surface weather map, based on weather station model information
- calculate the temperature gradient from either a spring day or winter day surface weather map in order to construct an explanation of the seasonal change in temperature variations
- collect data from weather maps for a particular location to complete a data table in order to support an explanation for how the movement and interactions of air masses and fronts result in changes in weather conditions (the data collected will include temperature, dew point, wind direction, wind speed, cloud cover, and air pressure)

In **Session 2: Individual Student Work**, students will use information from **Session 1** and satellite imagery. Students, working individually, will:

- synthesize information from surface and satellite weather maps to draw conclusions
- describe the complex interactions between air masses, which result in changes in weather conditions
- describe the influence of the ocean on the development of air masses and hurricanes
- communicate how the uneven heating of Earth’s surface and the prevailing global wind patterns drive the movement of air masses
- communicate how the uneven heating of Earth’s surface causes density differences in the atmosphere, which are responsible for circulation patterns of air pressure systems, the interactions of different air masses at frontal boundaries, and resulting weather phenomena
- explain the relationship between air temperature, atmospheric density, atmospheric pressure, and altitude using information from the pressure block models

Standards Alignment

Primary PE

HS-ESS2-8: Evaluate data and communicate information to explain how the movement and interactions of air masses result in changes in weather conditions.

Related PEs

HS-ESS1-7: Construct an explanation using evidence to support the claim that the (Moon, eclipses, tides, and) seasons change cyclically.

HS-ESS2-5: (Plan and conduct an investigation) of the properties of water and its effect on (Earth materials and) surface processes.

HS-ESS2-2: Analyze geoscience data to make the claim that one change to Earth’s surface can create feedbacks that cause changes to Earth’s systems.

Primary DCI

HS-ESS2.D: Weather and Climate: (NYSED) Concepts of density and heat energy can be used to explain observations of weather patterns.

Related DCIs

HS-ESS1.B: Earth and the Solar System: (NYSED) Earth and celestial phenomena can be described by principles of relative motion and perspective. (HS-ESS1-7)

HS-ESS2.C: The Role of Water in Earth’s Surface Processes: The abundance of liquid water on Earth’s surface and its unique combination of physical and chemical properties are central to the planet’s dynamics. These properties include water’s exceptional capacity to absorb, store, and release large amounts of energy, transmit sunlight, expand upon freezing, dissolve, transport materials, and lower the viscosities and melting points of rocks. (HS-ESS2-5)

HS-ESS2.D: Weather and Climate: The foundation for Earth’s global climate system is the electromagnetic radiation from the Sun, as well as its reflection, absorption, storage, and redistribution among the atmosphere, ocean, and land systems, and this energy’s re-radiation into space. (HS-ESS2-2)

Primary SEP

Obtaining, Evaluating, and Communicating Information: Communicate scientific ideas in multiple formats. (HS-ESS2-8) Students will evaluate meteorological data in small groups to identify and communicate relationships between weather variables such as air density and heat energy. Students will collaboratively plot hurricane paths, determine the location of frontal boundaries, calculate seasonal changes in temperature gradients, and use a model to communicate relationships between weather variables.

Related SEPs

Constructing Explanations (and Designing Solutions): Construct an explanation based on valid and reliable evidence obtained from a variety of sources (including students’ own investigations, models, theories, simulations, peer review) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future. (HS-ESS1-7) Students will use evidence from weather maps to explain how changes in weather are caused by the movement and interactions of air masses, global wind patterns, and uneven heating and cooling of Earth’s surface and oceans.

(Planning and) Carrying Out Investigations: Planning and carrying out investigations in 9-12 (builds on K-8 experiences and) progresses to include investigations that provide evidence for (and test) conceptual, mathematical, physical, and empirical models. (HS-ESS2-5) Students will use scientific ideas about the processes related to Earth’s systems to describe the movement and interactions of air masses and the resulting changes in weather conditions.

Analyzing and Interpreting Data: Analyze data using (tools, technologies, and/or) models (e.g., computational or mathematical) in order to make valid and reliable scientific claims (or determine optimal design solutions) (HS-ESS2-2). Students will use a pressure block model of columns of warm and cold air to explain relationships between air temperature, atmospheric density, atmospheric pressure, and altitude. Additionally, students will use data from various weather maps to make valid scientific claims about the movement of air masses, the interactions of air masses along frontal boundaries, and the influence of planetary winds on the movement and characteristics of hurricanes.

Using Mathematics and Computational Thinking: Use a computational representation of phenomena to describe and/or support claims and/or explanations. Students will identify numerical relationships among weather variables (including using data obtained from weather maps) that affect weather patterns and conditions locally and seasonally.

Developing and Using Models: Develop a model, based on evidence, to illustrate the relationship between systems or between components of a system. Students will construct a model to represent columns of air to illustrate the relationships between air temperature, atmospheric pressure, atmospheric density, and altitude.

Primary CCC

Patterns:

- Different patterns may be observed at each of the scales at which a system is studied and can provide evidence for causality in explanations of phenomena. Students will identify patterns in weather variables within a warm column of air and a cold column of air from using a model. Additionally, students will use patterns derived from station models to locate frontal boundaries. Students will be able to recognize patterns of prevailing global winds by plotting the path of a northern or a southern hemisphere hurricane.
- Empirical evidence is needed to identify patterns. Students will use data obtained from station models to identify the patterns of movement and interaction of air masses that result in changes in weather conditions.

Related CCCs

Structure and Function: The functions and properties of natural and designed objects and systems can be inferred from their overall structure, the way their components are shaped and used, and the molecular substructures of its various materials. (HS-ESS2-5) Students will identify and describe hydrologic cycle processes responsible for changes in atmospheric variables and resulting weather phenomena.

Stability and Change: Feedback (negative and positive) can stabilize or destabilize a system. (HS-ESS2-2) Students will identify and describe how changing atmospheric variables cause changes in global and synoptic atmospheric patterns that, in turn, create changing weather conditions.

Cause and Effect: Empirical evidence is required to differentiate between cause and correlation and make claims about specific causes and effects. (HS-ESS3-1) Students will make claims about how characteristics of ocean and land surfaces influence the development of air masses. Additionally, students will explain the cause-and-effect relationships of weather variables due to differences in air density that occur along frontal boundaries.

Stability and Change: Change and rates of change can be quantified and modeled over very short or very long periods of time (some changes are irreversible). Students will quantify spatial and temporal changes in weather conditions that occur seasonally and over a four-day period.

Teacher Directions

Preparing the Investigation:

This investigation, broken into **Session 1: Part 1**, **Session 1: Part 2**, and **Session 2**, is intended to satisfy a maximum of 100 minutes toward the 1200 minutes of hands-on laboratory experience required by Part 100 of the Commissioner’s Regulations. Investigations must be documented as satisfactory to be counted toward the 1200-minute requirement.

Appropriate safety practices are left to local discretion. The teacher should communicate the appropriate use and care of safety equipment and the associated procedures to students.

In **Session 1 (Part 1 and Part 2)**, students will collaborate in pairs.

- 1) Pair students.
- 2) Make sure the ***Student Directions***, ***Student Answer Packet 1***, and the ***Weather Investigation Packet*** have been copied and are ready for distribution to students.

In **Session 2**, students will individually complete ***Student Answer Packet 2*** questions and activities.

- 1) Make sure the ***Student Answer Packet 2*** has been copied and is ready for distribution to students.

Determine how many station setups you will need knowing that students will be working in pairs for **Session 1: Part 1** and **Part 2** of the investigation. For example, if you have 30 students, you will need 15 setups.

Materials Needed for One Setup: (It is recommended that these materials be compiled together in containers for reuse.)

- 1 set of “pressure blocks” (5 red and 5 blue. These blocks, red — $1\frac{3}{8}$ inch x $1\frac{3}{8}$ inch x $1\frac{3}{8}$ inch and blue — $\frac{11}{16}$ inch x $1\frac{3}{8}$ inch x $1\frac{3}{8}$ inch, need to be assembled prior to the investigation.) Templates have been provided on pages 8 and 9 for the construction of blocks from card stock.
- 2 index cards (3-inch x 5-inch) (labeled surface 1 and surface 2)
- Sharpened colored pencils, including blue and red (They will be used for plotting the path of the hurricanes and identifying locations of frontal boundaries. It is recommended that teachers have multiple individual sharpeners available for student use for precise line drawing during this investigation.)
- 1 calculator (Have additional calculators available so that each student has access to their own calculator to complete the investigation questions.)
- 1 hand lens (Have additional hand lenses available to read station models so that each student has access if needed.)

Materials per Student:

- ***Weather Investigation Packet***
- ***Student Directions***
- ***Student Answer Packet 1***
- ***Student Answer Packet 2***

Preparation for Session 1 (Part 1) of Investigation:

- 1) Assemble materials for the number of students in your class. (It is recommended that materials for each setup be compiled in containers ready for student use.)
- 2) Make sure the ***Weather Investigation Packet***, ***Student Directions***, and ***Student Answer Packet 1*** have been copied and are ready for distribution to each student.
- 3) Assign pairs of students.
- 4) Distribute the ***Weather Investigation Packet***, ***Student Directions***, and ***Student Answer Packet 1*** to each student. Though students are working in pairs, each student is responsible for completing the activities in ***Student Answer Packet 1*** individually. Some activities will require student pairs to complete different tasks and compare these tasks in embedded questions in the packet.
- 5) All materials should be collected at the end of **Part 1** *only* if **Part 2** will be completed on a different day.

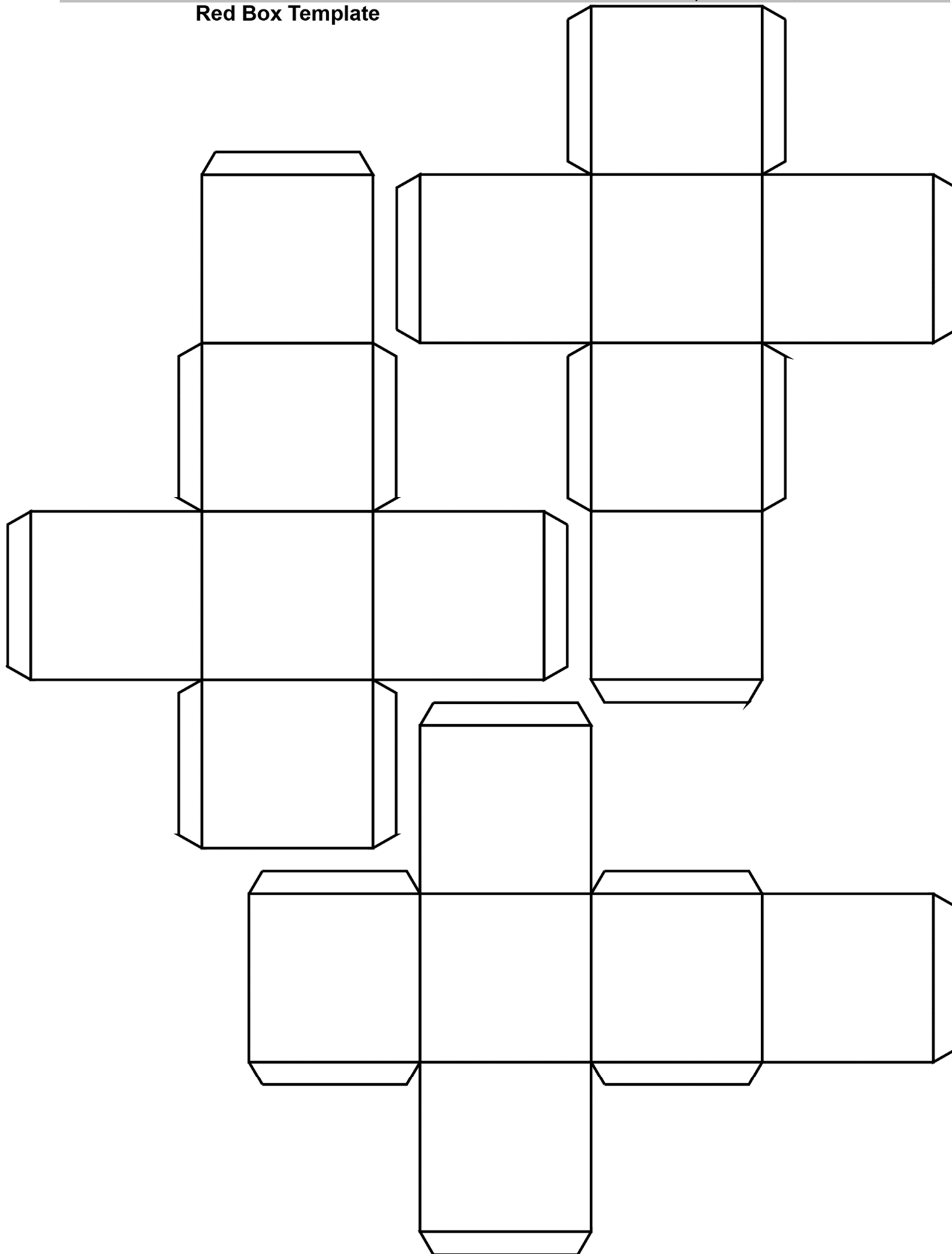
Preparation for Session 1 (Part 2) of Investigation:

- 1) Distribute all materials from **Part 1** back to each student.

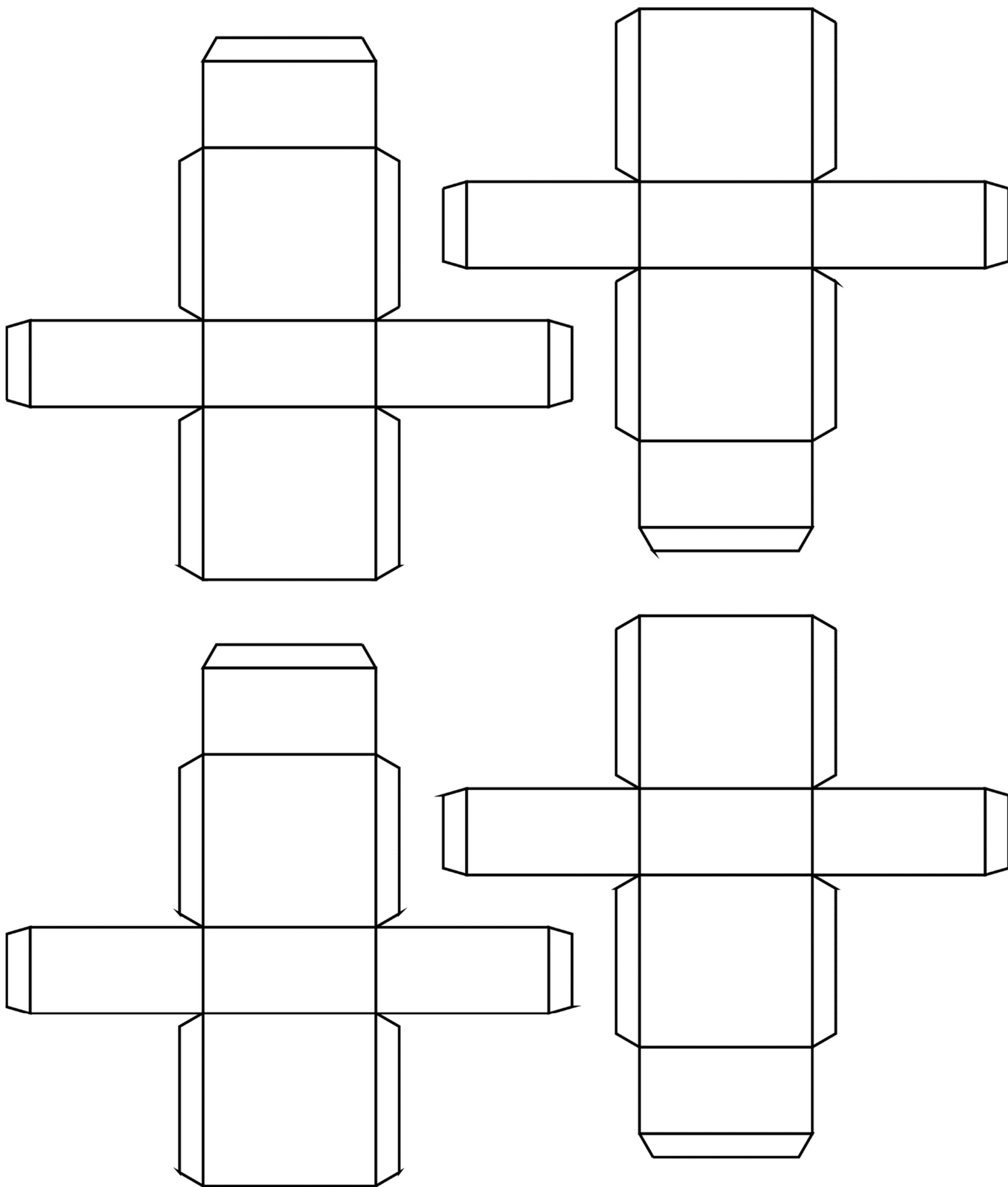
Preparation for Session 2 of Investigation:

- 1) Make sure ***Student Answer Packet 2*** has been copied and is ready for distribution to each student.
- 2) Return the ***Weather Investigation Packet***, ***Student Directions***, and ***Student Answer Packet 1*** to students, as they can be used during **Session 2** in which the investigation questions are answered.

Red Box Template



Blue Box Template



Student Directions with Teacher Annotations

Focus Questions: While performing this investigation, use the following focus questions to guide your work.

1. How can pressure block models of warm air and cold air be used to describe atmospheric density and pressure differences?
2. How do prevailing global winds drive the motions of hurricanes, tropical cyclones, and air masses?
3. How can locations of frontal boundaries be predicted based on data from weather maps?
4. How does latitude influence seasonal temperature gradients for fixed locations?
5. How can changes in weather conditions over three days be described using data collected from a sequence of weather maps?

Inform students that they are not answering the focus questions at this time. The focus questions are included as an overview of the concepts related to this investigation. These questions will be addressed through the students' performance of this investigation.

Procedure:

Part 1

1. Write your name on ***Student Answer Packet 1***. All answers for **Part 1** will be recorded in ***Student Answer Packet 1***. Obtain one set of five red/warm “pressure blocks,” one set of five blue/cold “pressure blocks,” and two index cards (3-inch x 5-inch).

Pressure blocks may be purchased or preassembled using red and blue card stock with the template provided.

2. You will use the red and blue “pressure blocks” to investigate a model representing the effect of air temperature and density on atmospheric pressure.

The blocks used in this investigation represent parcels of air. Red blocks represent warm air and blue blocks represent cold air.

All blocks are assumed to have the same:

- mass, regardless of their volume
- base size
- downward pressure exerted on the surface

3. Place one tall red block and one short blue block next to each other on their square bases on the surface of the table.

4. Place one more block of the same color on top of each block already on the table so there is a total of two blocks of each color in each stack, as shown in *Figure 1*.

Answer questions 4a and 4b.

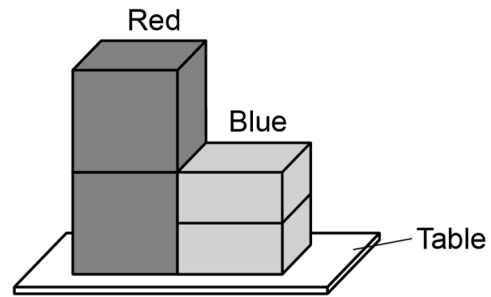


Figure 1

5. Remove one red block from the stack and place the index card labeled **surface 1** on top of the blocks.
6. Place two red and one blue block on top of **surface 1**, as shown in *Figure 2* (for a total of three blocks in each stack).

Answer questions 6a, 6b, and 6c.

Student setup should look like the diagram on the right.

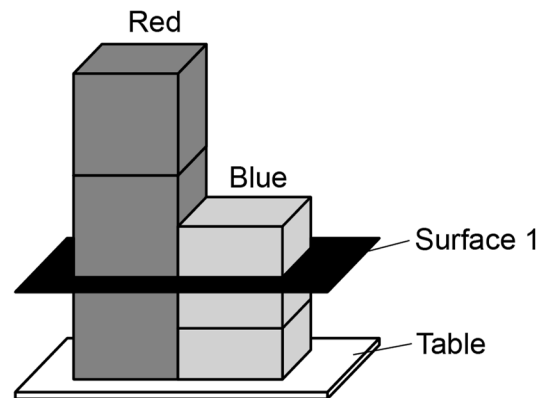


Figure 2

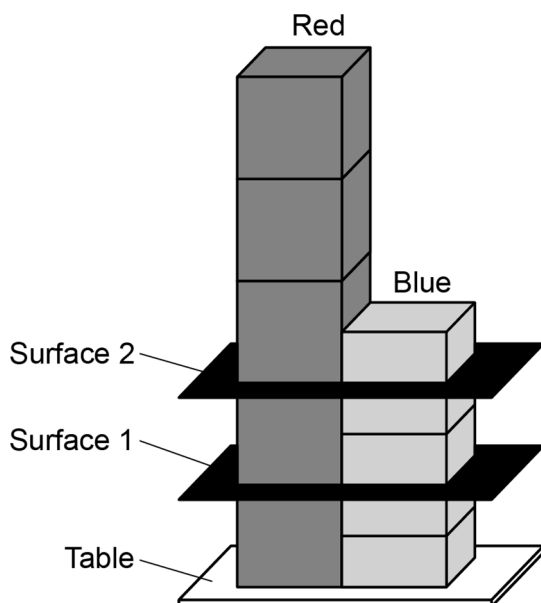


Figure 3

7. Remove one red block and add one blue block. Then, place the index card that is labeled **surface 2** on the top of the stack. Add three red blocks and one blue block so that there is a total of five red and five blue blocks separated at **surface 1** and **surface 2**, as shown in *Figure 3*.

Answer question 7.

Student setup should look like the diagram on the left.

8. Use the *Model of Stacked Blocks with Pressure Isolines* and the *Vertical Cross-Section of Air Pressure (at 7 a.m. December 9, 2010) in a Column of Warm and Cold Air* in the **Weather Investigation Packet** to answer questions 8a, 8b, and 8c.

Return the pressure block materials to your teacher.

Weather occurs at different temporal and spatial scales. In this investigation, two different meteorological scales are used: synoptic scale and global scale. Synoptic scale phenomena cover an area of several hundred to thousands of kilometers. High- and low-pressure systems shown on typical weather maps are synoptic in scale. Pressure and convection are important meteorological principles that are at the root of the development of fronts and large-scale weather systems such as hurricanes. Hurricanes occur globally in both hemispheres and are a result of pressure systems developing over tropical waters.

9. Each member of the group will plot the track of **one** hurricane. One group member will plot the track of Hurricane Andrew (with a blue pencil); the other group member will plot the track of Hurricane Orson (with a red pencil). Use the appropriate hurricane tracking chart and the data to plot the path of your selected hurricane found on pages 5 or 6 in **Student Answer Packet 1**. Connect your plots with a line using your colored pencil. Draw arrows along the path to indicate the direction of the hurricane.

The tables contain data for a northern hemisphere tropical system (Hurricane Andrew) and a southern hemisphere tropical cyclone (Hurricane Orson).

Data for Hurricane Andrew

Date	Time	Latitude (°N)	Longitude (°W)	Wind (mph)	Pressure (mb)	Storm Type
8/22/1992	6:00 a.m.	25.6	67.0	75	994	Category 1 hurricane
8/22/1992	6:00 p.m.	25.7	69.7	109	969	Category 2 hurricane
8/23/1992	6:00 a.m.	25.5	72.5	150	947	Category 4 hurricane
8/23/1992	6:00 p.m.	25.4	75.8	173	922	Category 5 hurricane
8/24/1992	6:00 a.m.	25.4	79.3	150	937	Category 4 hurricane
8/24/1992	6:00 p.m.	25.8	83.1	132	947	Category 4 hurricane
8/25/1992	6:00 a.m.	26.6	86.7	132	948	Category 4 hurricane
8/25/1992	6:00 p.m.	27.8	89.6	144	941	Category 4 hurricane
8/26/1992	6:00 a.m.	29.2	91.3	138	955	Category 4 hurricane
8/26/1992	6:00 p.m.	30.9	91.6	58	991	Tropical Storm
8/27/1992	6:00 a.m.	32.1	90.5	35	997	Tropical Depression
8/27/1992	6:00 p.m.	33.6	88.4	29	999	Tropical Depression

Data for Tropical Cyclone (Hurricane) Orson

Date	Time	Latitude (°S)	Longitude (°E)	Wind (mph)	Pressure (mb)	Storm Type
4/18/1989	6:00 a.m.	12.6	124.1	46	991	Category 1 hurricane
4/18/1989	6:00 p.m.	12.4	122.3	58	985	Category 1 hurricane
4/19/1989	6:00 a.m.	12.4	121.4	69	976	Category 2 hurricane
4/19/1989	6:00 p.m.	12.3	120.4	92	954	Category 3 hurricane
4/20/1989	6:00 a.m.	12.4	119.3	109	939	Category 3 hurricane
4/20/1989	6:00 p.m.	12.9	118.2	127	929	Category 4 hurricane
4/21/1989	6:00 a.m.	14.2	116.7	132	914	Category 4 hurricane
4/21/1989	6:00 p.m.	15.5	116.2	144	905	Category 5 hurricane
4/22/1989	6:00 a.m.	17.4	116.2	150	905	Category 5 hurricane
4/22/1989	6:00 p.m.	20.0	116.1	138	911	Category 4 hurricane
4/23/1989	6:00 a.m.	23.7	117.3	58	985	Category 1 hurricane

10. Compare the track that you plotted with the track plotted by your partner.

Answer question 10.

11. Use your knowledge of global wind patterns and hurricanes to complete the task in number 11.

Answer question 11.

Remind students to use their ***Weather Investigation Packet*** in addition to ***Student Answer Packet 1***.

Part 2

12. Mid-latitude cyclones are characterized by frontal boundaries. Each member of the group will use information from the station models on the *Surface Observation Map* found in ***Student Answer Packet 1*** to find the center of the low-pressure system and its associated fronts. Communicate with your partner to determine the location of the center of this mid-latitude cyclone. Both members will indicate this location with a capital “L” using the red colored pencil. Determine and draw (using appropriate frontal symbols) the location of **both** the warm front and the cold front associated with this low-pressure system. Draw the warm front using the red colored pencil and draw the cold front using the blue colored pencil.

Answer question 12.

Remind students to use the *United States Map* to help them identify the names of states as evidence for front locations.

13. Use the surface weather map in your ***Weather Investigation Packet*** from *Thursday, December 15, 2011*, to complete the task in number 13.

Answer question 13.

14. Energy is distributed unevenly throughout Earth’s atmosphere due to seasonal variations in the angle and intensity of sunlight. Determine the temperature variations between two fixed locations on Earth at different times of the year.

Answer question 14.

15. Weather forecasts can be produced by observing daily weather conditions over periods of time. The sequence of surface weather maps in the ***Weather Investigation Packet*** shows weather data for December 14, 15, and 16, 2011.

Use weather data from the maps to answer question 15.

Session 2

Return the ***Weather Investigation Packet***, ***Student Directions***, and ***Student Answer Packet 1*** to the students, as they can be used to answer some ***Student Answer Packet 2*** questions.

Your task in **Session 2** is to apply what you have learned in **Session 1** to independently answer the investigation questions found in ***Student Answer Packet 2***.

Answer/Scoring Key

Student Answer Packet 1

4a. Acceptable responses include, but are not limited to:

- Pressure is greater/increases.

4b. Acceptable responses include, but are not limited to:

- Because red blocks and blue blocks have the same mass and red blocks have a larger volume, red blocks represent air with a lower density.
- Warmer air is less dense than cold air. Therefore, the mass in the warm air/red blocks is distributed over a larger volume.

6a. Acceptable responses include, but are not limited to:

- The downward pressure created by two red blocks will be greater than the downward pressure created by one blue block on **surface 1**.
- The pressure of one blue block will be less than two red blocks on **surface 1**.

6b. Acceptable responses include, but are not limited to:

- The downward pressure created by one red block will be less than the downward pressure exerted by the two blue blocks.
- The two blue blocks will have a greater pressure than one red block.

6c. Acceptable responses include, but are not limited to:

- The downward pressure exerted by three red blocks is the same as the downward pressure exerted by three blue blocks on the table.
- Both columns will exert the same downward pressure on the surface of the table since there are the same number of blocks in each column and all blocks have the same mass.

7. Acceptable responses include, but are not limited to:

- The downward pressure exerted by the red stack on **surface 2** is greater than the downward pressure exerted by the blue stack.
- The downward pressure of the red stack is three times greater than the downward pressure of the blue stack on **surface 2**.

8a. Florida: Acceptable responses include any value from 12,100 to 12,300 meters.

Massachusetts: Acceptable responses include any value from 11,300 to 11,500 meters.

Note: Responses must include the appropriate units.

8b. Acceptable responses include, but are not limited to:

- The rate at which pressure changes with altitude in a warm air column is lower than the rate at which pressure changes in a cold air column.
- The rate of change in the warm air column is less than the rate of change in the cold air column.

8c. Acceptable responses include, but are not limited to:

- The warm air column, like in Florida, is thicker/taller than the cold air column, like in Massachusetts, so the tropopause is at a higher altitude in Florida than in Massachusetts.
- Lower latitudes have warmer air columns that reach higher altitudes, causing the tropopause to be at a higher altitude. Higher latitudes, like in Massachusetts, generally have colder air columns, so the tropopause is at a lower altitude.

10. Acceptable responses include, but are not limited to:

- The track of Hurricane Andrew generally moved southeast to northwest until Andrew made landfall north of the Gulf of Mexico. Then, Andrew moved northeast. Hurricane Orson generally moved east to west, and then Orson turned south/southeast.
- Both hurricanes began their tracks moving east to west. However, Hurricane Andrew finished its track by moving northeast while Hurricane Orson moved south/southeast.
- Hurricane Andrew moved northeast after it traveled north of 30° north latitude, while Hurricane Orson moved south. Before that, both storms moved east to west.

11. Acceptable responses include, but are not limited to:

Northern hemisphere wind belt:

- northeast trades
- northeast winds

Southern hemisphere wind belt:

- southeast trades
- southeast winds

12. Acceptable responses include, but are not limited to:

Warm front:

- It is 66°F in Georgia, while it is 50°F in Tennessee, so there must be a front between these two locations.
- It is 38°F in Kansas, while it is 70°F in Arkansas south of it.
- It is 48°F in North Carolina, but it is 77°F in Florida, right below.

Cold front:

- Oklahoma is 63°F, while to the west it is 31°F in North Texas.
- Central Texas is 80°F, while west of it in Texas it is 47°F.

13. Acceptable responses include, but are not limited to:

- There are decreasing air pressure values on the map. The lowest pressure value shown is 996 mb, therefore the center of the low must be located at the lowest value.
- The isobar interval on the map is 4 millibars, and the values of the isolines decrease to the lowest value of 996 millibars north of Lake Huron in Canada.

14. Responses must identify the correct temperature gradient values shown below:

December 14, 2011 temperature gradient: $0.034^{\circ}\text{F}/\text{mi}$

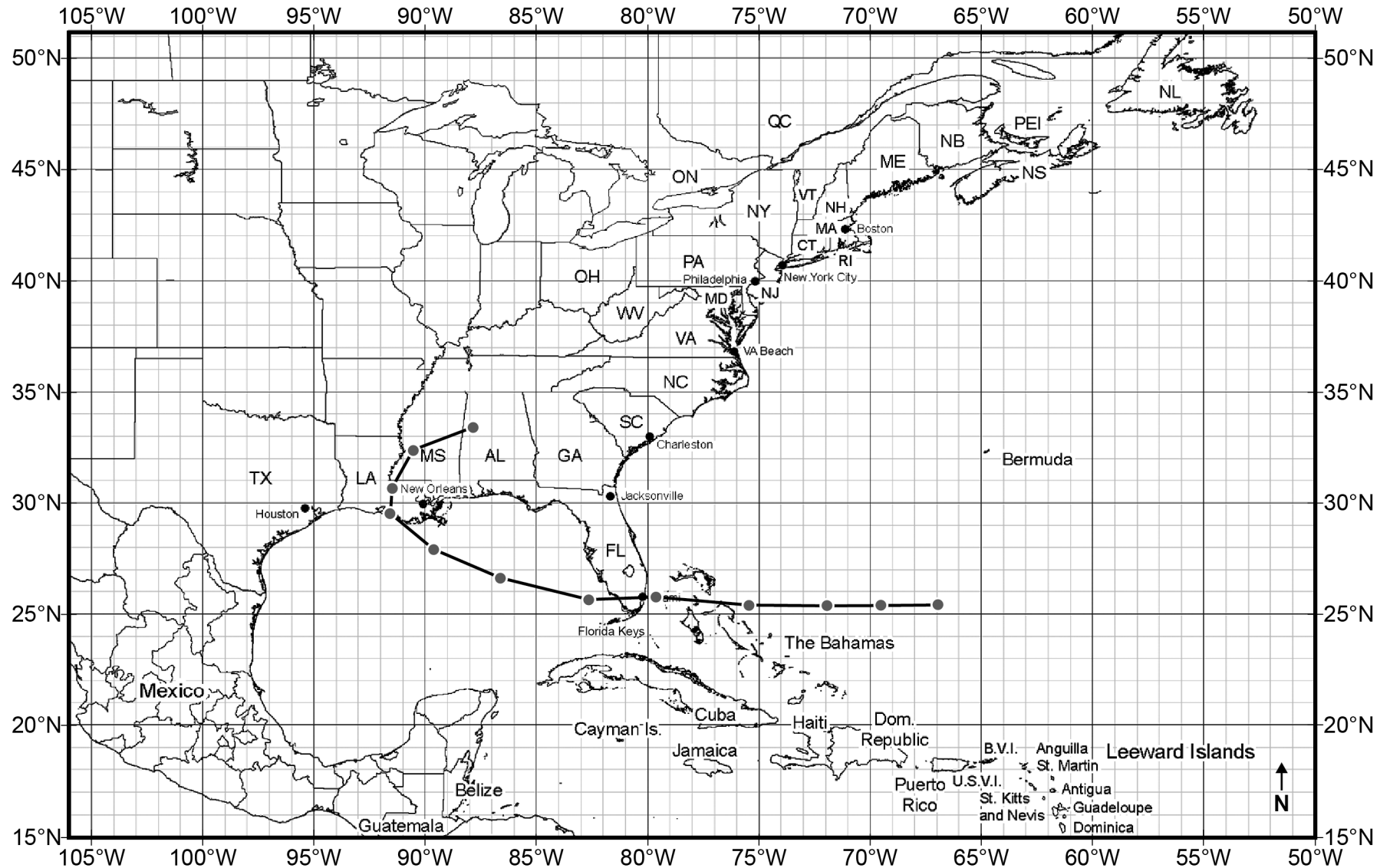
June 16, 2012 temperature gradient: $0.018^{\circ}\text{F}/\text{mi}$

15.

Date	Temperature (°F)	Dew Point (°F)	Wind Direction	Wind Speed (knots)	Cloud Cover (%)	Air Pressure (mb)
12/14/11	36	32	None	0	100	1029.8
12/15/11	43	38	None	0	100	1018.0
12/16/11	38	27	W or WNW	20	0	1013.0

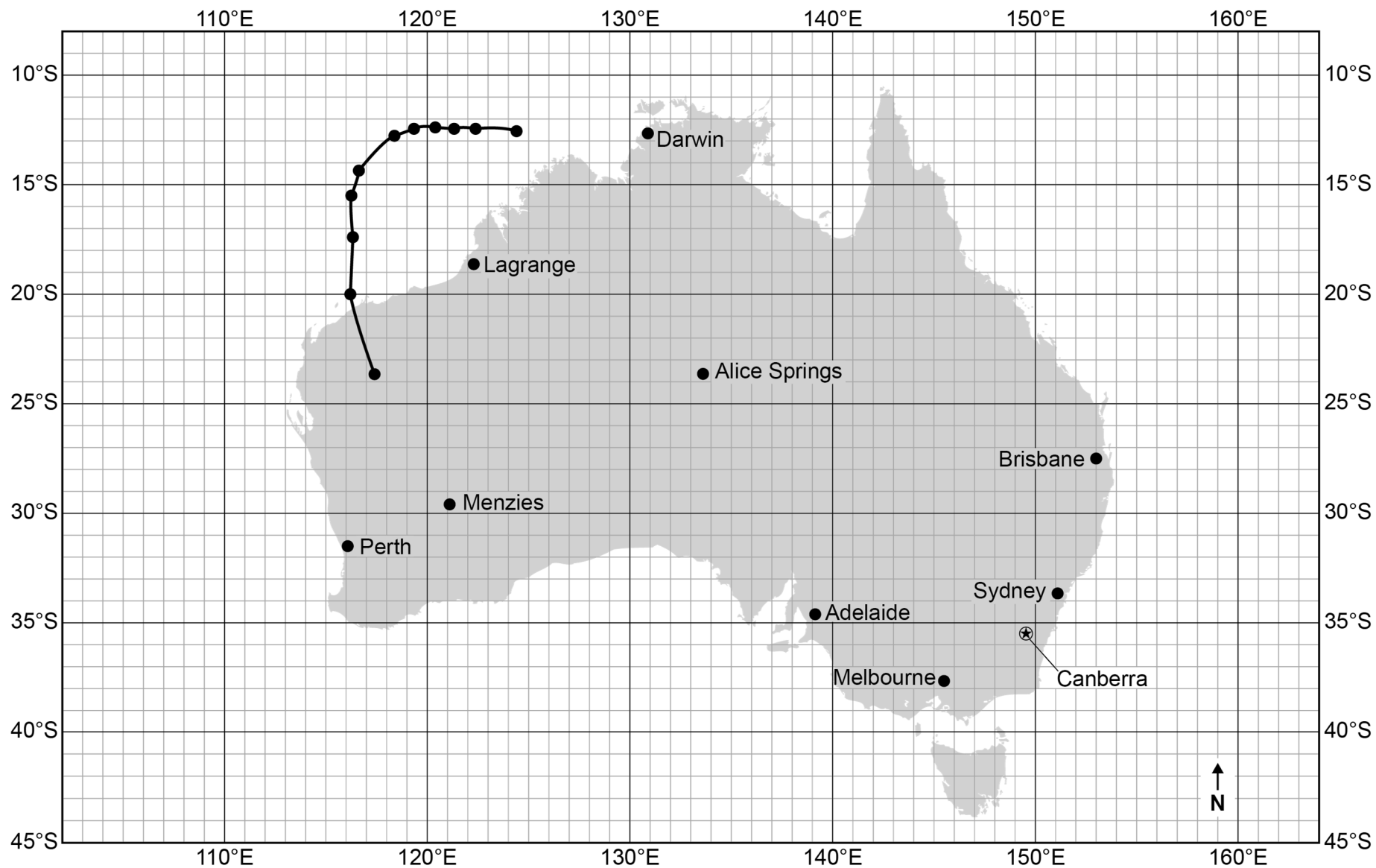
Name: _____

Hurricane Andrew Tracking Chart



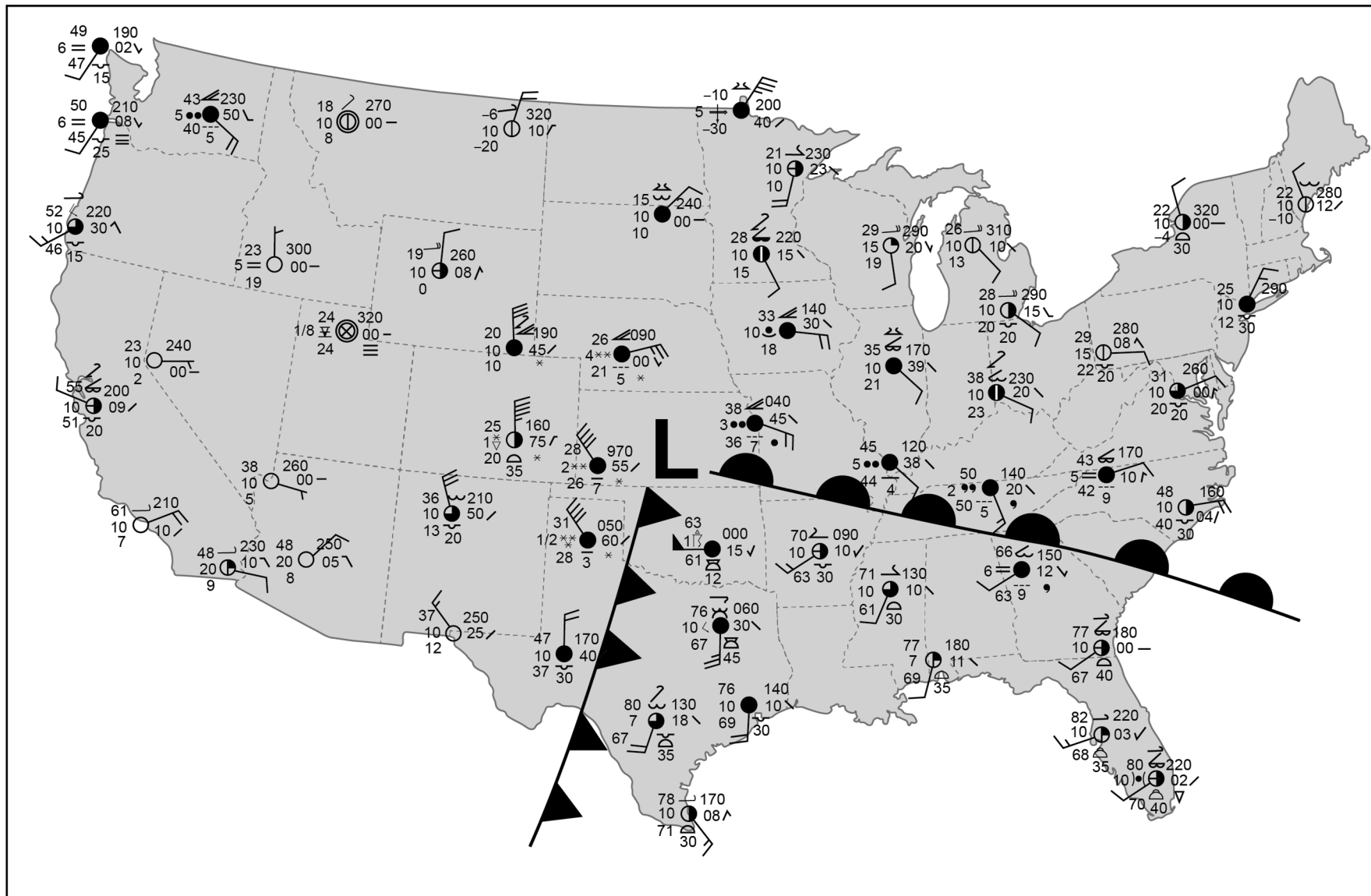
Name: _____

Tropical Cyclone (Hurricane) Orson Tracking Chart



Name: _____

Surface Observation Map



Answer/Scoring Key

Student Answer Packet 2

1. Acceptable responses include, but are not limited to:
 - Red and blue blocks have the same mass, but blue blocks have a smaller volume.
 - The masses are identical, but the red blocks have lower densities because the volume is twice as large as the blue blocks.
2. Acceptable responses include numerical values that indicate a lowering of air pressure with an increase in altitude, for example:
 - At sea level in Massachusetts, the air pressure is approximately 1000 mb, and at an altitude of 10,000 meters, the air pressure is approximately 250 mb, so as altitude increases, air pressure decreases.
 - As altitude increases, air pressure decreases. The data that supports this is at the surface in Florida, the air pressure is approximately 1000 mb, and at an altitude of 7500 meters, the air pressure is approximately 400 mb.
- 3a. Acceptable responses include, but are not limited to:
 - On December 14, the air temperature in Abilene was 65°F and the air pressure was 1011.7 mb; after the cold front passed, the air temperature was 49°F and the air pressure increased to 1018.1 mb.
 - The air temperature went from 65°F to 49°F, while the air pressure changed from 1011.7 mb to 1018.1 mb due to the passage of the cold front.
 - When the cold front passed, it caused the air temperature to change from 65°F to 49°F, while the air pressure increased from 1011.7 mb to 1018.1 mb.

3b. Acceptable responses include, but are not limited to:

- Areas of precipitation moved closer to the east coast (away from Iowa and toward Ohio and surrounding states) as the low-pressure system moved northeast.
- Temperatures increased in mid-Atlantic states as the warm front moved toward the northeast.
- As the cold front moved to the east, air pressures increased in the mid-western states.
- In front of the cold front, the winds were from the south; behind the cold front, the winds were from the west and northwest.

4. Name of global wind belt:

- Midlatitude westerlies/prevaling westerlies/westerlies

Acceptable responses include, but are not limited to:

Description:

- Global winds located between 30° N latitude and 60° N latitude move from southwest to northeast, moving the low-pressure system in the same direction.
- During this time period, westerly winds moved the weather systems from west to east.
- Weather systems moved from west to east because global winds between 30° N and 60° N move from southwest to northeast.

5. Acceptable responses include, but are not limited to:

Hurricane Andrew:

- counterclockwise and inward

Tropical Cyclone (Hurricane) Orson:

- clockwise and inward

6. Acceptable responses include, but are not limited to:

- The ocean currents and ocean water temperatures on the west coast of the United States are too cold to allow storms to strengthen into a hurricane.
- Hurricanes require warm ocean water to form, and cold ocean currents are located in the Pacific Ocean off the west coast of the United States.

7. Acceptable responses include, but are not limited to:

Selected date: 12/14/11 Location: Canada Evidence: 1. air temperature is below freezing 2. under an area of precipitation 3. cloud cover is 100% 4. the dew point and air temperature are very close	Selected date: 12/15/11 Location: Northern Minnesota Evidence: 1. station model indicates snow as type of precipitation 2. air temperature is below freezing 3. wind direction is from the northwest 4. the dew point and air temperature are very close
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8a. Acceptable responses include, but are not limited to:

- Warmer, less dense air at a frontal boundary will rise, expand, cool, and condense to create clouds and precipitation.
- Colder, more dense air sinks at a frontal boundary, pushing warmer, less dense air aloft.

8b. Acceptable answers include, but are not limited to:

- The cold air behind the cold front is too dense and could not rise, expand, cool, and condense to form clouds and precipitation, while warm, less dense air ahead of the cold front does rise, expand, cool, and condense, forming precipitation.
- The colder air behind the cold front is more dense and sinks, preventing clouds from forming. But the warm air ahead of the cold front rises, expands, cools, and condenses, forming clouds and precipitation.

9. Acceptable responses include, but are not limited to:

- The December temperature gradient is greater than the June temperature gradient because the polar jet stream dips south of North Dakota but not to Texas.
- The June temperature gradient is less than the December temperature gradient because the jet stream is farther north of both locations in June.

10a. Acceptable responses include, but are not limited to:

- On December 14th, the temperatures were cool (mid-30°F) and the weather was dry; on December 15th; a warm air mass (50°F) arrived, bringing rain and warm air; on December 16th, a cold air mass arrived, resulting in cold temperatures (upper-30°F) and dry weather.

10b. Acceptable responses include, but are not limited to:

- The location over which an air mass forms determines the characteristics of the air mass. For example, an air mass that forms over the Gulf of Mexico will be warm and have moisture.
- Air masses that form over tropical land acquire the characteristics of warm and dry.